Pheromone Trail Formation at the Edge of Chaos: Phase Diagrams of a Lattice Ant Model

Abstract

We study a lattice-based ant model in which mobile agents deposit pheromone on a shared environment field, follow pheromone gradients with directional momentum, and explore stochastically. By sweeping parameters we construct phase diagrams revealing three regimes: frozen trail networks, transient trail dynamics (edge of chaos), and spatial disorder. A composite order parameter $C = S \times (1 - A)$—spatial structure times temporal decorrelation—locates the edge-of-chaos regime. We present three model variants in a progression toward biological realism. The *original model* uses an exploration noise parameter that creates a sharp composite peak (a true edge-of-chaos ridge) but "cheats"—with some probability, an ant ignores all signals and teleports to a random neighbour. The *simple model* replaces noise with a follow probability p_f, giving each ant a coherent meander-or-follow rule, but fails to produce a sharp phase transition because individual momentum creates structure even without social coupling. The *susceptibility model* makes the coupling state-dependent: all ants deposit pheromone unconditionally (biological secretion), but only follow trails when local pheromone concentration exceeds a detection threshold. This creates positive feedback bistability—following reinforces trails which recruit more followers—producing a sharp recruitment phase boundary. However, this boundary is a first-order-like collapse, not an edge-of-chaos ridge: the composite drops monotonically across it (block variance and autocorrelation decline together) rather than peaking. The composite instead peaks in the *erosion-driven transience* regime—where all ants follow, but high evaporation constantly reshapes the trail network ($C = 0.625$)—which is a genuine edge-of-chaos phenomenon along the persistence axis. The susceptibility model thus *decouples* two axes of order that the noise parameter conflated: **recruitment** (a sharp phase transition but not edge of chaos) and **persistence** (edge of chaos through erosion). The original model's single ridge was a superposition of these two distinct phenomena. Attempts to achieve edge of chaos through intrinsic ant mechanisms (habituation, sensory saturation, Hill function chemoreception) all fail because their noise is *correlated* with the pheromone field—only *uncorrelated* perturbations (external noise, environmental decay) can decouple spatial structure from temporal persistence.

1 The Model

1.1 Setup

Consider N agents (ants) on a $W \times H$ toroidal grid with Moore neighbourhood. Each cell is either empty or occupied by one ant. A continuous environment field $\phi(x, y, t) \geq 0$ represents pheromone concentration.

Each ant maintains a heading vector $\mathbf{h} = (h_r, h_c)$, updated as an exponential moving average of recent move directions.

1.2 Movement rule

At each time step, every ant attempts to move to an empty neighbouring cell. The probability of choosing neighbour j is:

$$p(j) = \frac{w(j)}{\sum_k w(k)}, \qquad w(j) = \underbrace{(\beta + \phi_j)^n}_{\text{pheromone attraction}} \times \underbrace{\max\bigl(0.01,\ 1 + \mu \cos\theta_j\bigr)}_{\text{momentum}} \tag{1}$$

where ϕ_j is the pheromone at cell j, β is a baseline exploration constant, n is the bias exponent, μ is the momentum strength, and θ_j is the angle between the ant's heading $\mathbf{h}$ and the direction to cell j.

With probability ε (the *noise* parameter), the ant ignores both pheromone and momentum, choosing uniformly among empty neighbours.

After each move, the heading is updated:

$$\mathbf{h} \leftarrow (1 - \alpha)\,\mathbf{h} + \alpha\,\Delta\mathbf{r} \tag{2}$$

where $\Delta\mathbf{r}$ is the move direction and α is the heading adaptation rate.

1.3 Pheromone dynamics

Each ant deposits a fixed amount δ of pheromone at its current cell every step. The field then undergoes two global operations:

$$\text{Evaporation:} \quad \phi \leftarrow (1 - \lambda)\,\phi \tag{3}$$

$$\text{Diffusion:} \quad \phi \leftarrow \phi + \sigma\,\nabla^2\phi \tag{4}$$

where λ is the evaporation rate and σ is the diffusion rate. The Laplacian $\nabla^2\phi$ is computed as the sum over Moore neighbours minus 8 times the centre value, normalised appropriately.

1.4 Parameters

Table 1 lists the full parameter set. The first group (topology, ant density, deposition) defines the physical setup. The second group (movement rule) controls individual ant behaviour. The third group (field operations) controls the shared pheromone medium.

Table 1: Model parameters. The two swept parameters are shown in bold.

Parameter	Symbol	Value	Role
Grid size	$W \times H$	100×100	Toroidal lattice
Ant density	ρ	2.5–5%	Fraction of cells occupied
Deposition	δ	5.0–6.0	Pheromone deposited per ant per step
Bias exponent	n	2.5–3.0	Superlinear pheromone preference
Bias base	β	0.1–0.3	Spontaneous exploration baseline
Momentum	μ	2.0–5.0	Directional persistence strength
Heading rate	α	0.3–0.4	EMA smoothing of heading
Noise	$\boldsymbol{\varepsilon}$	**0.0–0.50**	**Exploration probability**
Evaporation	$\boldsymbol{\lambda}$	**0.02–0.45**	**Pheromone decay rate**
Diffusion	σ	0.002–0.005	Pheromone spreading rate

2 Emergent Trail Formation

From uniform random initial placement, ants self-organise into trail networks through positive feedback: deposited pheromone attracts other ants, which deposit more pheromone, reinforcing the trail. Momentum causes ants to travel *along* trails rather than clustering at high-pheromone spots, producing linear highways rather than blobs.

Figure 1 shows trail formation in the equilibrium regime ($\varepsilon = 0$, $\lambda = 0.08$). By $t = 1000$, a persistent highway network has formed. Figure 2 shows the final ant positions alongside the pheromone field, revealing the correspondence between ant traffic and trail structure.

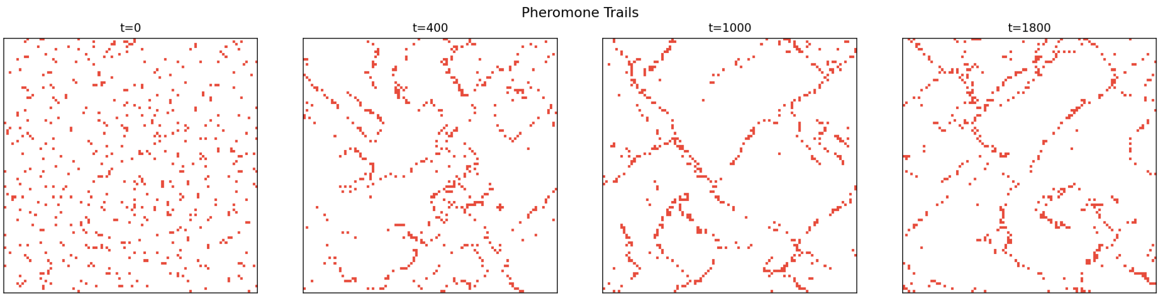

Figure 1: Ant positions over time in the equilibrium regime. From uniform random placement ($t = 0$), ants self-organise into persistent trail networks ($t = 1800$). Parameters: $n = 3$, $\beta = 0.1$, $\mu = 5$, $\varepsilon = 0$, $\lambda = 0.08$, $\sigma = 0.002$.

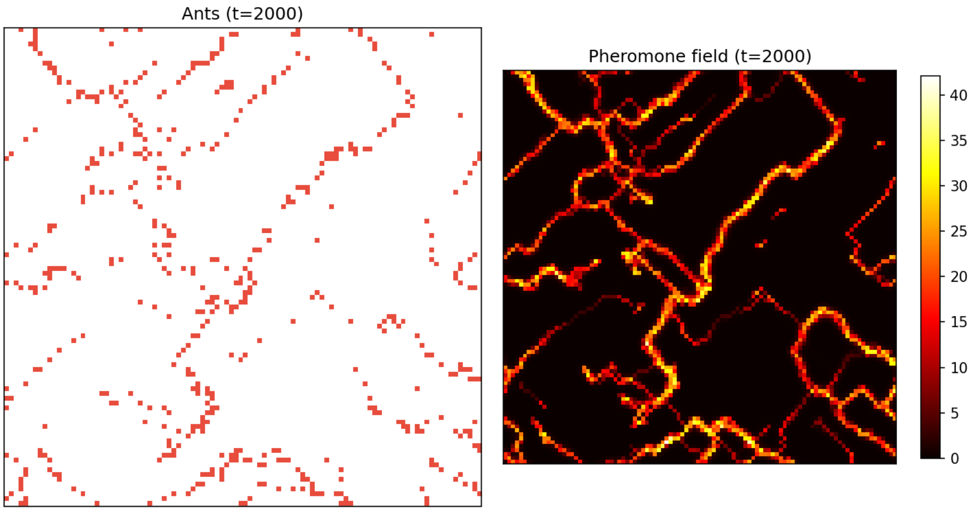

Figure 2: Equilibrium state at $t = 2000$. Left: ant positions. Right: pheromone field intensity. The trail network is a self-reinforcing structure maintained by ant traffic.

In the transient regime ($\varepsilon = 0.10$, $\lambda = 0.12$), trails form but continuously reorganise (Figure 3). This is the edge-of-chaos regime: structured but impermanent.

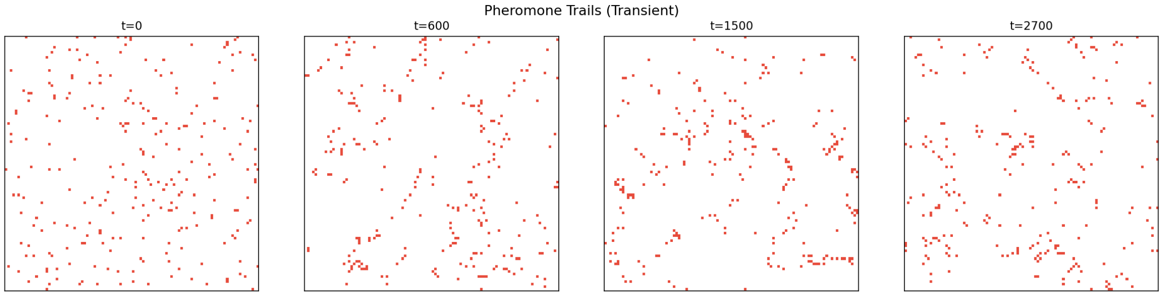

Figure 3: Transient regime: trails form and dissolve over time. The trail network at $t = 600$ differs from the network at $t = 2700$. Parameters: $n = 2.5$, $\beta = 0.3$, $\mu = 2$, $\varepsilon = 0.10$, $\lambda = 0.12$, $\sigma = 0.003$.

3 Order Parameters

To locate the edge of chaos, we need scalar quantities that distinguish the three regimes.

3.1 Field autocorrelation

The Pearson correlation between the pheromone field at times t and $t - \tau$:

$$A(\tau) = \frac{\sum_{x,y}(\phi_t - \bar{\phi}_t)(\phi_{t-\tau} - \bar{\phi}_{t-\tau})}{\sqrt{\sum(\phi_t - \bar{\phi}_t)^2}\sqrt{\sum(\phi_{t-\tau} - \bar{\phi}_{t-\tau})^2}} \tag{5}$$

This measures trail persistence: $A \approx 1$ for frozen trails, $A \approx 0$ for transient or disordered states. It distinguishes frozen from non-frozen but cannot distinguish "different trails" from "no trails"—both give low autocorrelation.

3.2 Trail linearity

Define "hot" cells as those where $\phi > 0.3 \cdot \max(\phi)$. Trail linearity is the fraction of hot cells with exactly 2 hot Moore neighbours:

$$L = \frac{|\{(x, y) : \mathrm{hot}(x, y) \ \wedge \ |\mathcal{N}_{\mathrm{hot}}(x, y)| = 2\}|}{|\{(x, y) : \mathrm{hot}(x, y)\}|} \tag{6}$$

Cells with exactly 2 hot neighbours form *chains*—the topological signature of a trail. Cells with many hot neighbours are blob-like; cells with 0–1 are isolated. High linearity indicates trail-like structure regardless of whether those trails persist over time.

3.3 The linearity metric and its limitations

Trail linearity was our initial structure metric and is used in the original and simple model sweeps (Sections 4–5). However, it has a fundamental limitation: it cannot distinguish trail structure produced by *collective following* from spatial texture produced by *individual momentum*.

When ants deposit pheromone while walking with directional persistence ($\mu > 0$), evaporation sculpts the deposition pattern into a landscape of hot spots. Some of these spots will randomly have exactly 2 hot Moore neighbours—scoring as "trail-like"—even though no trail-following behaviour produced them. This creates a linearity floor of $L \approx 0.20$–0.30 regardless of whether ants are following pheromone or walking independently.

In a metric comparison across susceptibility regimes (Section 6), linearity varied by only ∼10% between full trail-following ($\mathrm{ff} = 1.0$, $L = 0.25$) and zero following ($\mathrm{ff} = 0$, $L = 0.23$). The metric is essentially blind to the collective dynamics we are trying to measure.

3.4 Block variance: a statistical alternative

To replace linearity, we need a metric that captures spatial structure at a scale larger than individual cells. Trail networks create *mesoscale heterogeneity*: some regions of the grid contain highways with concentrated pheromone, while others are empty corridors. Random deposition from individually moving ants creates cell-level texture but is approximately uniform at the mesoscale.

We divide the $W \times H$ grid into non-overlapping blocks of size $b \times b$ and compute the mean pheromone in each block. The block variance is the normalised variance of these block means:

$$B = \frac{\mathrm{Var}(\bar{\phi}_{\mathrm{blocks}})}{\bar{\phi}^2} \tag{7}$$

where $\bar{\phi}_{\mathrm{blocks}}$ is the vector of block-averaged pheromone values and $\bar{\phi}$ is the global mean. Dividing by $\bar{\phi}^2$ makes the metric scale-invariant with respect to overall pheromone intensity.

Block variance is high when pheromone is concentrated in certain regions (trail networks) and low when it is spread uniformly (random deposition). In our metric comparison, block variance dropped by $\sim$50% between the following and non-following regimes—far more discriminating than linearity's $\sim$10%.

3.5 Composite order parameter

Neither observable alone peaks at the edge of chaos. Autocorrelation is high in the frozen regime and low everywhere else. The structure metric (linearity L or block variance B) is high in the frozen and transient regimes. We construct:

$$C = S \times (1 - A) \tag{8}$$

where S is the structure metric (L for the original and simple models, B for the susceptibility model).

This is maximised where trails *exist* ($S > 0$) but are *transient* ($A < 1$), and is zero in two failure modes: no trails ($S = 0$) or frozen trails ($A = 1$). The product is structurally similar to the LMC statistical complexity $C_{\mathrm{LMC}} = H \times D$ of López-Ruiz, Mancini & Calbet (1995), where our structure metric S plays the role of disequilibrium D (spatial order) and $(1 - A)$ plays the role of entropy H (temporal disorder).

4 Phase Diagram: The Original Model

We swept noise $\varepsilon \in [0, 0.50]$ and evaporation $\lambda \in [0.02, 0.45]$ on a 13×13 grid (169 simulations), each running 600 steps on a 60×60 grid with $\rho = 3.75\%$. All other parameters were held fixed at $n = 2.5$, $\beta = 0.3$, $\mu = 2.0$, $\alpha = 0.4$, $\delta = 6.0$, $\sigma = 0.005$.

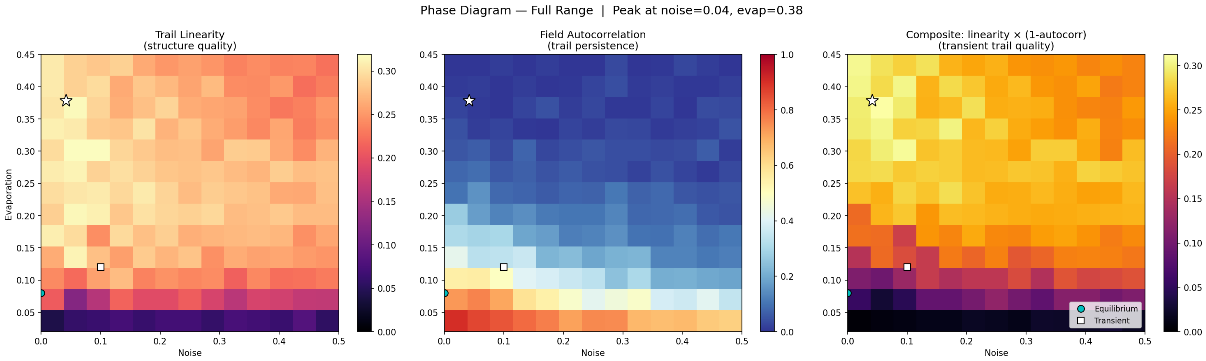

Figure 4: Phase diagram of the original model. Left: trail linearity. Centre: field autocorrelation. Right: composite $C = L \times (1 - A)$. Star marks the composite peak ($\varepsilon = 0.04$, $\lambda = 0.38$, $C = 0.312$). Circle: equilibrium spec. Square: transient spec.

The phase diagram (Figure 4) reveals three regimes:

- **Frozen trails** (bottom strip, $\lambda < 0.10$): Persistent highway networks. Autocorrelation $A > 0.5$. The composite is low because $(1 - A) \approx 0$.
- **Transient trails** (middle-to-upper band): Trails form but rewire. Autocorrelation $A \in [0.05, 0.3]$. Linearity remains high. The composite peaks here.
- **Disorder** (top-right, high ε and λ): No coherent structure. Both L and A are low.

The composite peak at $\varepsilon = 0.04$, $\lambda = 0.38$ reveals that the edge of chaos has very low noise (ants are mostly deterministic followers) and very high evaporation (the environment rapidly erases trails). The non-equilibrium regime is driven by *environmental erasure* rather than *behavioural randomness*.

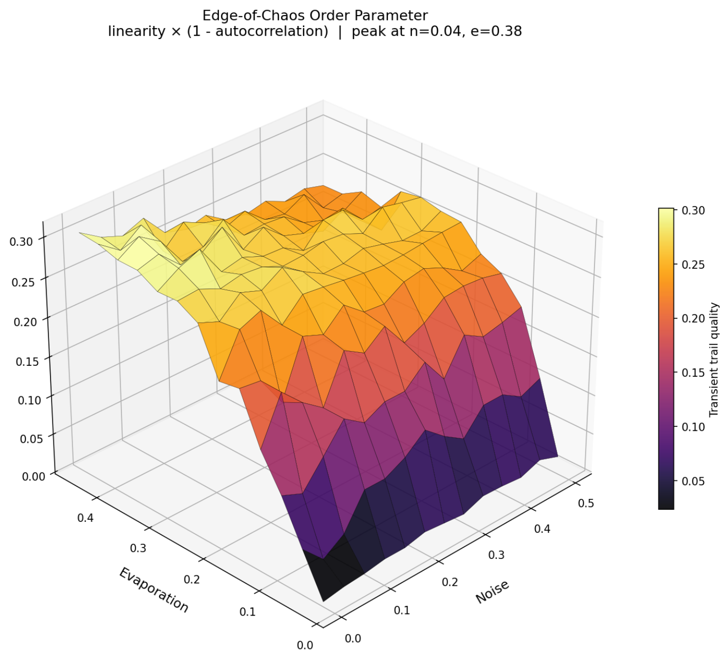

Figure 5: 3D surface of the composite order parameter. The ridge runs along low noise at moderate-to-high evaporation. It drops to zero in two directions: downward (trails freeze) and rightward (structure destroyed by noise).

The composite surface (Figure 5) forms a ridge, not a peak—it extends along the evaporation axis, reflecting the asymmetry between the two disruption mechanisms. Evaporation destroys trail *persistence* without destroying trail *structure*; noise destroys both.

4.1 Regimes across parameter space

Figures 6 and 7 show the pheromone fields and ant positions at representative points across the noise–evaporation plane. Reading left to right (increasing noise) shows the transition from frozen highways to disorder. Reading bottom to top (increasing evaporation) shows trails becoming more transient and eventually dissolving. The edge-of-chaos regime—visible as bright, structured trails with moderate ant clustering—occupies a diagonal band from low noise / high evaporation to moderate noise / low evaporation.

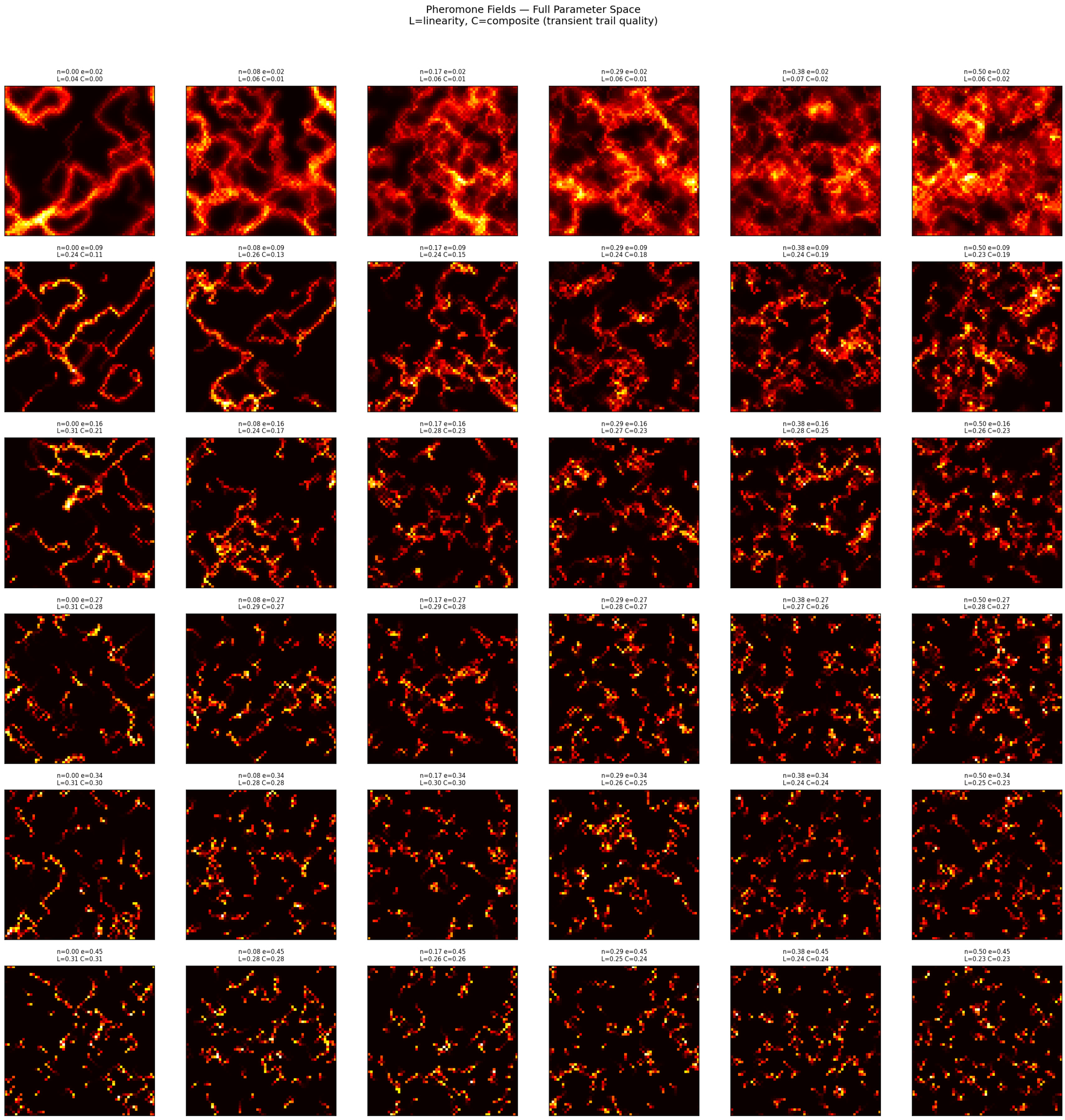

Figure 6: Pheromone fields across the noise–evaporation parameter space. Rows increase in evaporation (bottom to top); columns increase in noise (left to right). Bottom-left: frozen trail networks with high pheromone concentration. Top-right: spatial disorder with no coherent structure. The diagonal band between these extremes shows transient trails—the edge of chaos. L = linearity, C = composite order parameter.

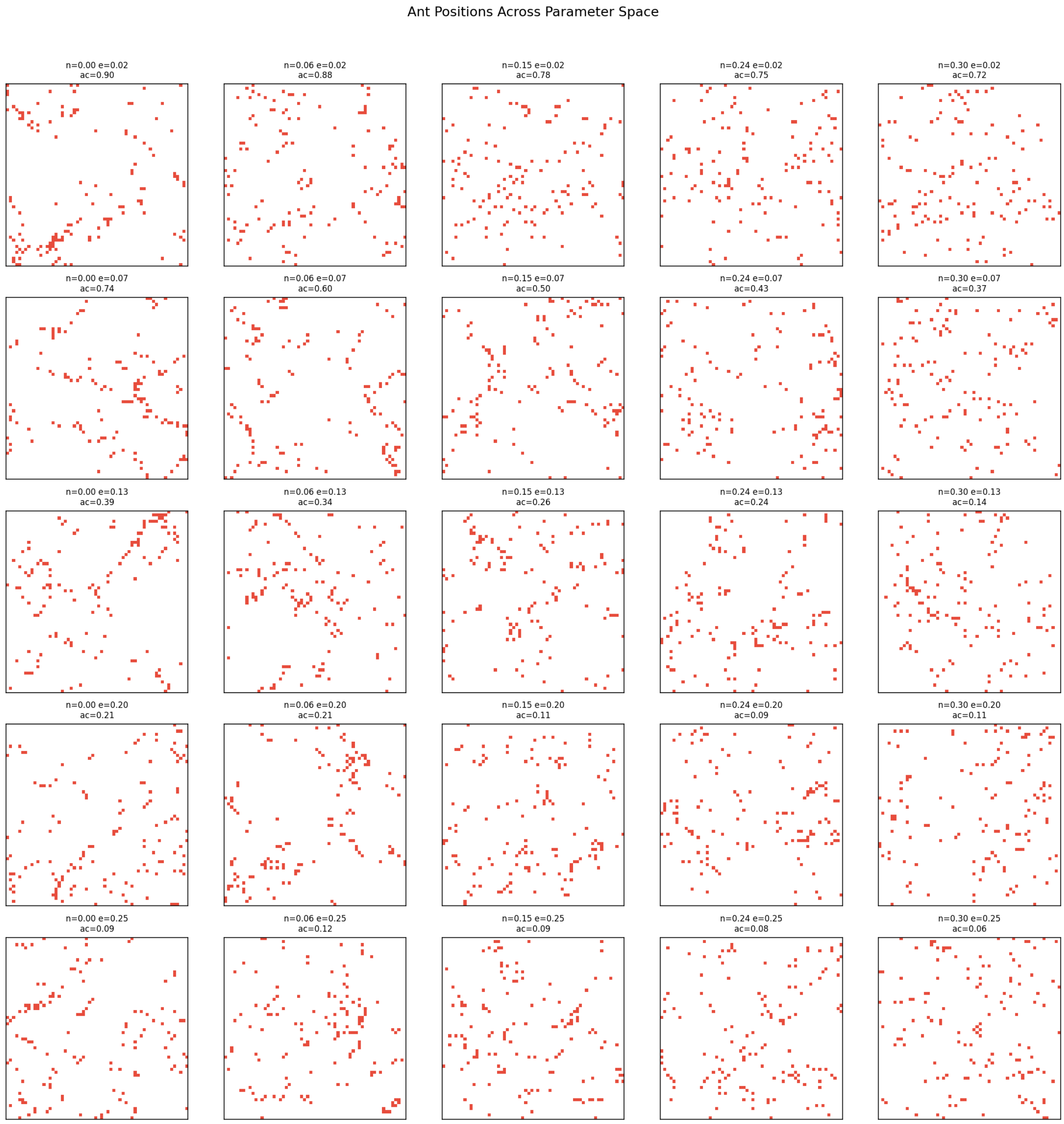

Figure 7: Ant positions corresponding to Figure 6. In the frozen regime (bottom-left), ants cluster tightly along persistent trail networks. In the transient regime (middle), ants form loose linear aggregations. In the disordered regime (top-right), ants are uniformly scattered with no spatial structure.

5 The Simple Model: Can We Do Without Noise?

5.1 Motivation

The noise parameter ε is a brute-force intervention: with probability ε, the ant ignores *everything*—pheromone, momentum, heading—and moves to a uniformly random neighbour. This is not a behavioural description. No real ant has a subroutine that occasionally overwrites its entire decision-making with uniform randomness.

We wanted a model where an individual ant has a simple, coherent behaviour—walk with gentle persistence, occasionally respond to a chemical signal—and complexity emerges purely from social interaction through the pheromone medium. The individual should be boring; the collective should be complex.

5.2 Formulation

Replace the six-parameter movement rule (1) with a coin flip:

$$\text{Each step:} \quad \begin{cases} \text{with probability } p_f: & \text{pick neighbour weighted linearly by } \phi_j + 0.01 \\ \text{otherwise:} & \text{move randomly with momentum } \mu \end{cases} \tag{9}$$

When following pheromone, the weighting is linear (no exponent n, no baseline β). When not following, the ant walks with its intrinsic momentum—it tends to keep going in its current direction, like a real ant exploring.

This gives three parameters:

Table 2: Simple model parameters.

Parameter	Symbol	Role
Follow probability	p_f	How often an ant reads the social signal
Evaporation	λ	How fast the shared memory decays
Diffusion	σ	How far the signal spreads spatially

Momentum μ and heading rate α are structural constants—properties of ant locomotion, not parameters of the information system.

5.3 Sweep results: high momentum ($\mu = 3.0$)

We swept $p_f \in [0, 1]$ and $\lambda \in [0.02, 0.45]$ with $\sigma = 0.005$ fixed.

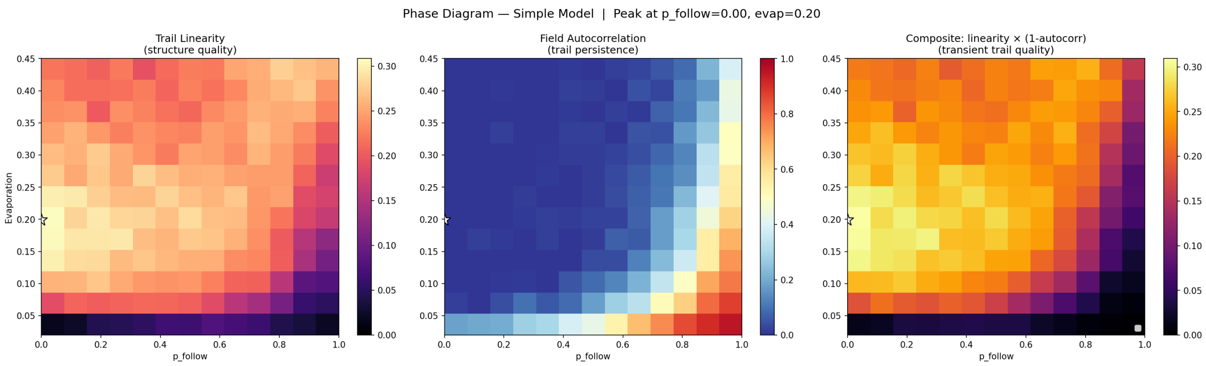

Figure 8: Phase diagram of the simple model with momentum $\mu = 3.0$. The composite peaks at the corner: $p_f = 0$, $\lambda = 0.20$, $C = 0.309$. There is no interior peak—no edge of chaos.

The composite peaks at $p_f = 0$ (Figure 8)—meaning the "best" transient trails come from ants that *never follow pheromone*. This is a boundary artifact, not an edge of chaos.

The problem: with $\mu = 3.0$, the momentum term in the non-following branch weights forward movement $4\times$ vs sideways and $\sim 300\times$ vs backward. Ants walk in near-straight lines, creating stream-like spatial structure without any pheromone feedback. The linearity metric saturates at $L \approx 0.30$ regardless of p_f.

The fundamental asymmetry: The original model's noise ε destroys both pheromone following *and* momentum simultaneously. The simple model's p_f only controls the pheromone signal while leaving momentum intact. There is no disorder regime—you cannot reach $L = 0$ by adjusting p_f alone when momentum creates order on its own.

5.4 Low momentum ($\mu = 0.5$): the fix

At $\mu = 0.5$, forward movement is only 1.5× vs sideways. Ants meander gently—not Brownian, but not streaming.

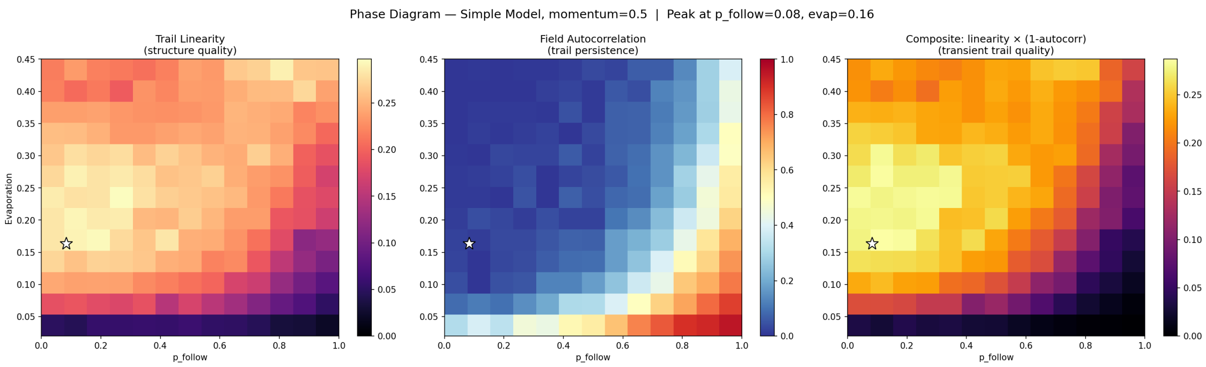

Figure 9: Phase diagram of the simple model with momentum $\mu = 0.5$. The composite peak moves to the interior: $p_f = 0.08$, $\lambda = 0.16$, $C = 0.287$.

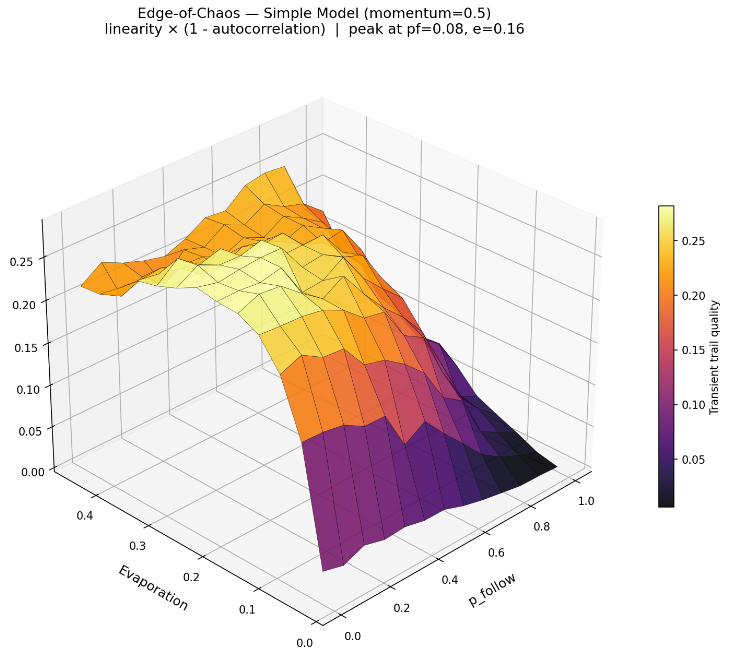

Figure 10: Composite surface for the simple model at $\mu = 0.5$. A shallow hill appears, but the peak is pressed against the $p_f = 0$ edge.

The composite now has an interior peak at $p_f = 0.08$, $\lambda = 0.16$ (Figures 9–10). But the peak is pressed hard against the boundary—$p_f = 0.08$ means ants follow pheromone only 8% of the time.

The linearity floor at $p_f = 0$ dropped from ~ 0.30 (at $\mu = 3.0$) to ~ 0.24, but this is still high enough that the composite is dominated by the autocorrelation factor.

5.5 Diffusion as the third axis

We swept all three parameters: $p_f \times \lambda \times \sigma$ in a $9 \times 8 \times 7$ grid (504 simulations) at $\mu = 0.5$.

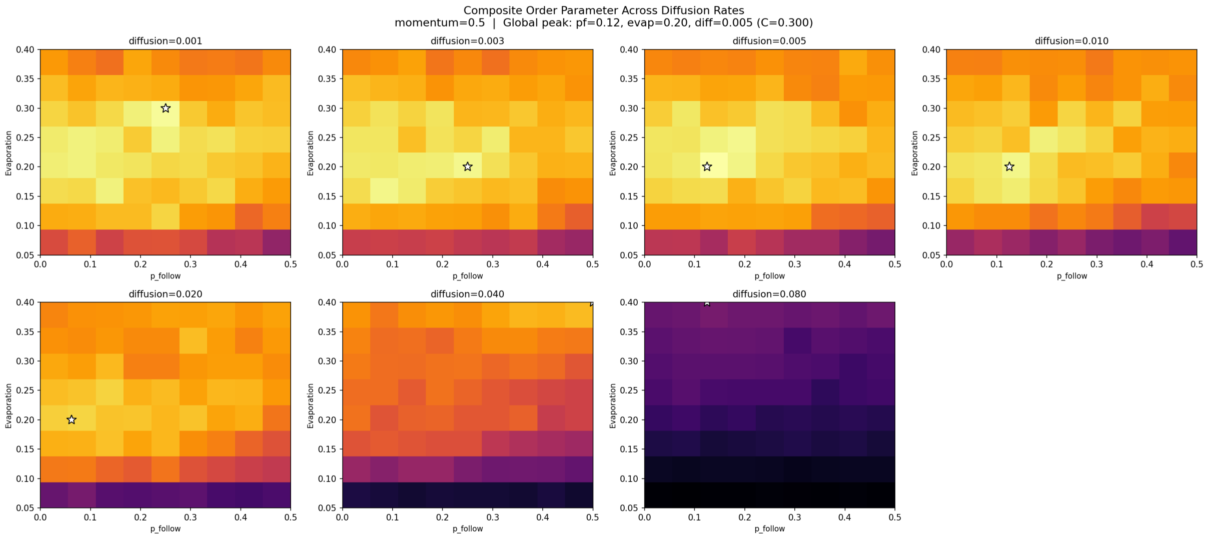

Figure 11: Composite heatmaps at each diffusion level. Stars mark the peak within each slice. Global peak: $p_f = 0.12$, $\lambda = 0.20$, $\sigma = 0.005$, $C = 0.300$. At $\sigma = 0.08$, pheromone smears into fog and linearity collapses.

The global peak sits at $p_f = 0.12$, $\lambda = 0.20$, $\sigma = 0.005$ ($C = 0.300$). Diffusion controls the spatial scale of the pheromone signal:

- **Too low** ($\sigma < 0.003$): pixel-thin trails; ants must be directly on them to detect pheromone. Higher p_f needed to compensate.
- **Optimal** ($\sigma \approx 0.005$): trails create a detectable gradient $\sim$2 cells wide. The ant's effective "sensing range."
- **Too high** ($\sigma > 0.04$): trails blur into a uniform haze; linearity collapses to $L \approx 0.08$.

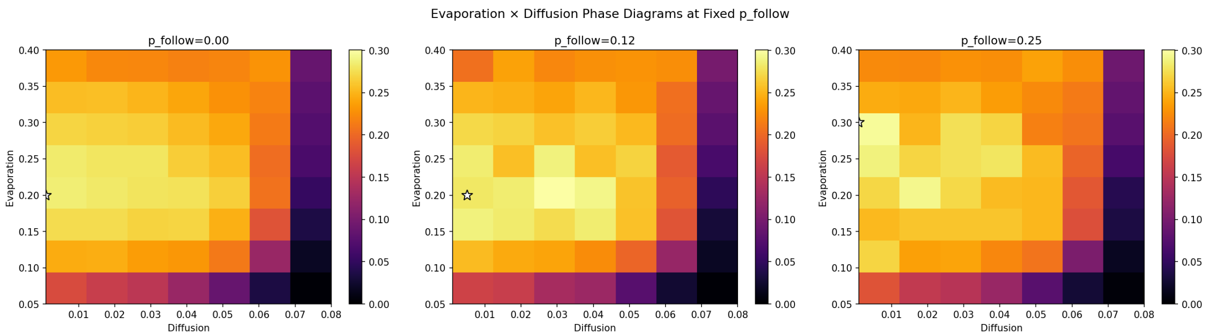

Figure 12: Evaporation vs diffusion at three fixed p_f values. A diagonal band of good composites runs from (low λ, low σ) to (high λ, high σ): faster-spreading pheromone must evaporate faster to avoid becoming fog.

The evaporation–diffusion interaction (Figure 12) shows a diagonal band: the two parameters are coupled through the physics of the pheromone field but are not redundant. Diffusion is genuinely a third independent axis controlling spatial scale, while evaporation controls temporal scale and p_f controls coupling strength.

5.6 Momentum $\mu = 0.6$

At $\mu = 0.6$ (forward weighting $1.6\times$ vs sideways), the peak moves to $p_f = 0.042$, $\lambda = 0.16$, $C = 0.301$—even closer to the $p_f = 0$ edge.

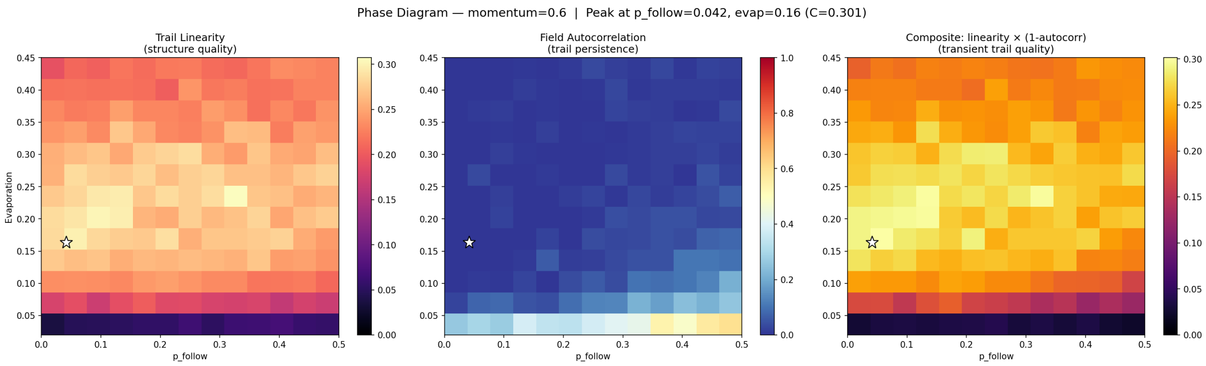

Figure 13: Phase diagram at $\mu = 0.6$. The peak at $p_f = 0.042$ confirms the trend: any momentum above ~ 0.3 creates enough spatial structure from random deposition to saturate the linearity metric.

This confirms the pattern: increasing momentum pushes the composite peak toward $p_f = 0$, because the random-walk branch produces progressively more trail-like structure on its own.

6 The Susceptibility Model: State-Dependent Following

6.1 Motivation

The simple model's p_f fails because each ant's decision is independent of the field's current state. Can we achieve sharp transitions through state-dependent coupling instead?

Real ants secrete pheromone continuously as they walk—a chemical byproduct, not a deliberate signal—but only *follow* trails when pheromone concentration exceeds a detection threshold. This creates feedback: whether an ant follows depends on the pheromone field, which itself depends on whether ants are following.

6.2 Formulation

All ants deposit pheromone unconditionally every step (biological secretion). Movement is state-dependent:

$$\text{Each step:} \quad \begin{cases} \text{if } \max_{j \in \mathcal{N}} \phi_j > s: & \text{follow pheromone with momentum} \\ \text{otherwise:} & \text{walk with momentum only} \end{cases} \tag{10}$$

In both branches, the ant maintains directional persistence (momentum μ). When following, neighbour weights are $w_j = (\phi_j + 0.01) \times (1 + \mu \cos \theta_j)$: a linear pheromone attraction modulated by the heading direction. This causes ants to walk *along* trails rather than converging on pheromone peaks, producing linear highways. When not following, weights are purely momentum-based: $w_j = \max(0.01, 1 + \mu \cos \theta_j)$, producing a gentle meander.

The parameter s is the *susceptibility threshold*—the pheromone concentration an ant must detect in its Moore neighbourhood to begin following. The key parameters are:

Table 3: Susceptibility model parameters.

Parameter	Symbol	Role
Susceptibility threshold	s	Pheromone level needed to trigger following
Evaporation	λ	Pheromone decay rate
Diffusion	σ	Pheromone spreading rate

Unlike p_f, the susceptibility threshold s does not directly set the fraction of ants that follow. Instead, the follow fraction is an *emergent* quantity determined by the interplay between deposition, evaporation, and the threshold.

6.3 The positive feedback mechanism

The susceptibility model creates **bistable dynamics** through positive feedback: pheromone exceeds threshold → ants follow → more deposition on trail → pheromone stays above threshold (**lock-in**); conversely, pheromone below threshold → no following → evaporation erodes it → pheromone stays below (**no trail**). This creates a sharp phase boundary between "all recruited" and "none recruited," determined by the balance of deposition rate, evaporation, and susceptibility threshold. Unlike the noise model's external destruction, criticality here arises from the system's own feedback dynamics.

6.4 Sweep results

We swept 17 susceptibility values ($s \in [0.05, 20]$), 16 evaporation values ($\lambda \in [0.05, 0.50]$), and 4 diffusion rates ($\sigma \in \{0.002, 0.005, 0.010, 0.020\}$), totalling 1,088 simulations on an 80×80 grid with $\mu = 2.0$, $\delta = 5.0$, and 800 steps. The composite uses block variance: $C = B \times (1 - A)$.

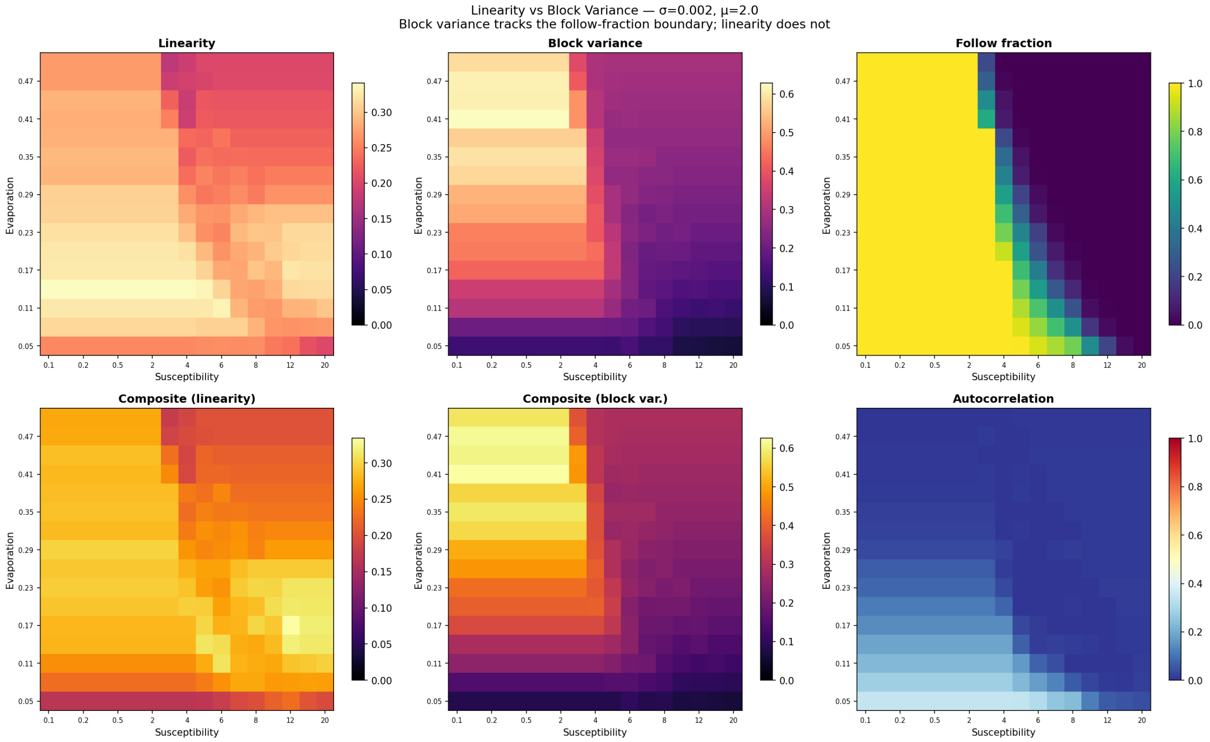

Figure 14: Linearity vs block variance as structure metrics. Top row: linearity, block variance, and follow fraction. Bottom row: composites formed with each metric, and autocorrelation. Linearity is nearly flat across regimes (∼10% variation). Block variance tracks the follow-fraction boundary, dropping by ∼50% when following ceases.

Figure 14 compares the two metrics: linearity is flat across regimes; block variance tracks the follow-fraction boundary.

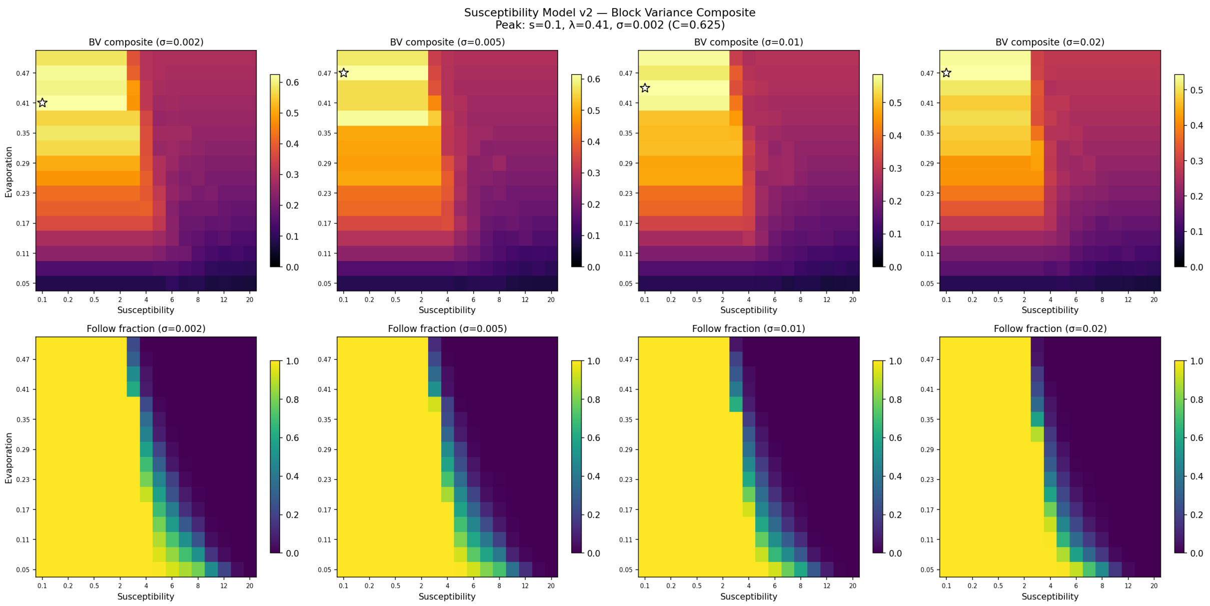

Figure 15: Block variance composite and follow fraction across four diffusion rates. Top row: composite $C = B \times (1 - A)$. Bottom row: follow fraction. The composite peak (star) sits in the high-evaporation, low-susceptibility regime where all ants follow but evaporation constantly reshapes the trail network—erosion-driven transience. Higher diffusion shifts the follow-fraction boundary rightward (pheromone spreads further, lowering the effective detection threshold).

The phase diagram (Figure 15) reveals three regimes:

- **Frozen** (low s, low λ): Nearly 100% of ants follow. Persistent highway networks with high autocorrelation and high block variance.
- **Transient** (moderate s, moderate λ): Partial recruitment. The follow fraction transitions sharply from high to zero. Trails form, attract followers, strengthen, get eroded, collapse, and reform elsewhere.
- **No following** (high s or high λ): Follow fraction drops to zero. Only unconditional deposition remains; block variance is low.

The follow-fraction boundary is a sharp diagonal—a genuine phase transition produced by positive feedback, not the smooth crossover seen in the p_f model. The transition curves (Figure 16) are nearly vertical sigmoids, steepening at low evaporation where positive feedback is strongest.

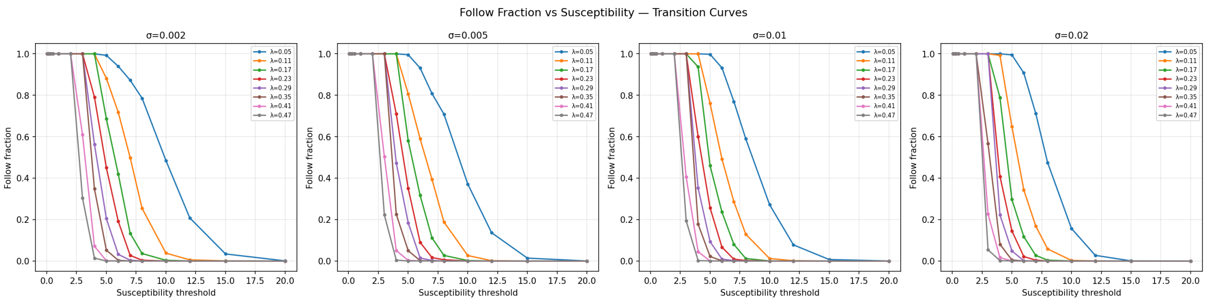

Figure 16: Follow fraction vs susceptibility threshold at each diffusion rate, coloured by evaporation. The transition steepens at low evaporation (stronger positive feedback).

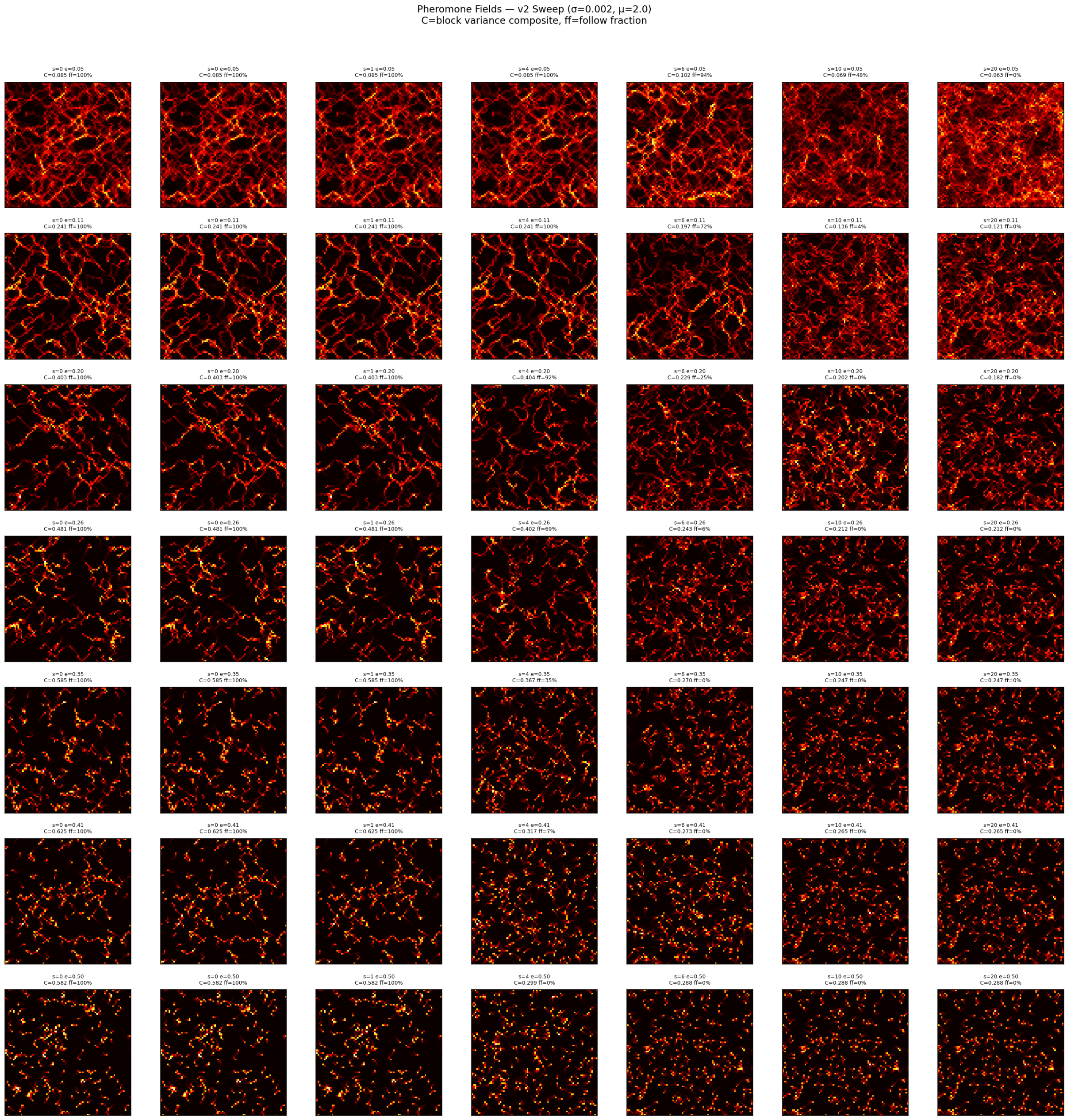

Figure 17: Pheromone fields across the susceptibility–evaporation plane. Low susceptibility (left): bright highway networks from near-total recruitment. High susceptibility (right): uniform dim glow from unconditional deposition alone. The transition from trails to disorder is visually abrupt, matching the sharp follow-fraction boundary.

7 Study 4: Decoupling Recruitment and Persistence

The composite peaked not at the recruitment boundary but in the erosion-driven transience regime ($s = 0.1$, $\lambda = 0.41$, $f = 1.0$, $C = 0.625$). This section investigates why.

7.1 Why the standard composite misses the recruitment boundary

The composite $C = B \times (1 - A)$ rewards configurations that are simultaneously *structured* (high B) and *transient* (low A). At the recruitment boundary, B is declining (fewer followers means weaker trails) even though A is also low. In the all-following regime at high evaporation, B remains maximal while A drops—yielding a higher product. The composite therefore peaks not at the recruitment boundary but in the erosion-driven regime.

7.2 A recruitment-sensitive composite

To target the recruitment phase boundary specifically, we introduce a recruitment weighting factor:

$$C_R = B \times (1 - A) \times 4f(1 - f) \tag{11}$$

where f is the follow fraction. The factor $4f(1-f)$ is maximised at $f = 0.5$ (half the ants following) and vanishes at $f = 0$ and $f = 1$, forcing the composite to peak at the recruitment transition rather than deep in the ordered or disordered regime.

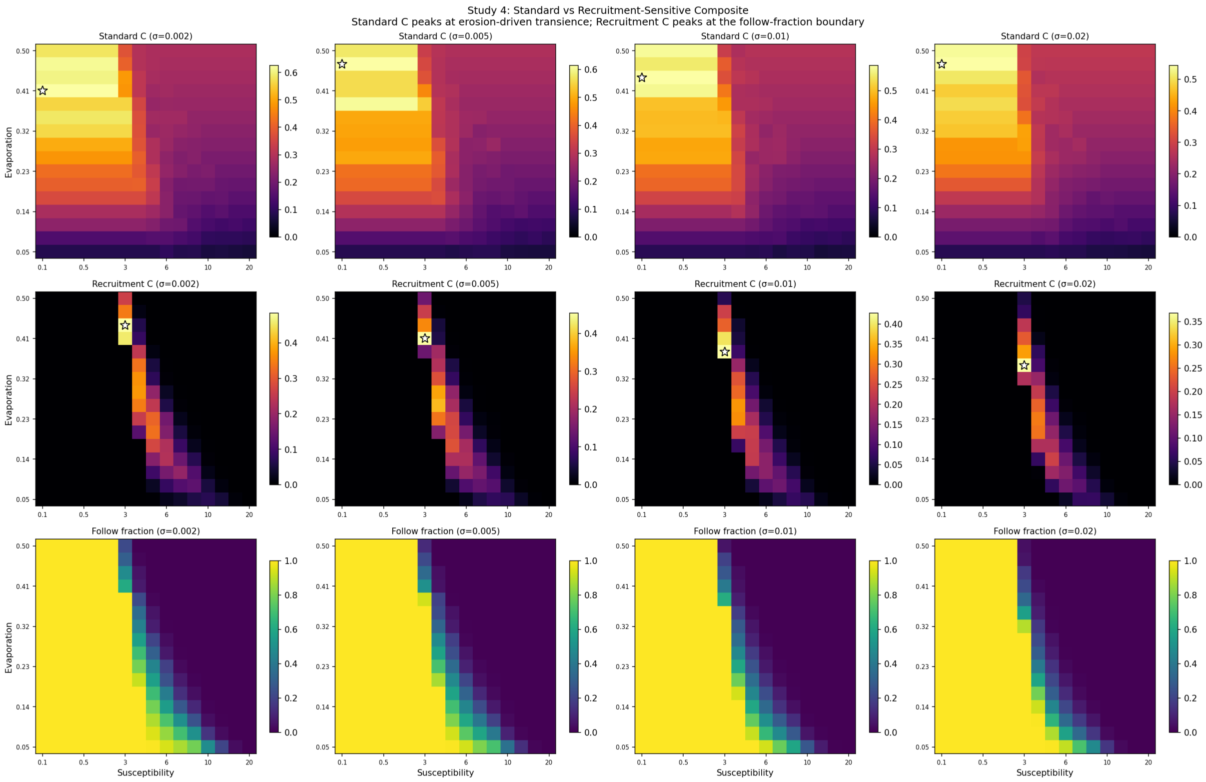

Figure 18: Standard vs recruitment-sensitive composite across four diffusion rates. Top row: standard $C = B \times (1 - A)$, peaking in the all-following regime (top-left, star). Middle row: recruitment composite $C_R = B \times (1 - A) \times 4f(1 - f)$, peaking at the follow-fraction boundary (star on diagonal). Bottom row: follow fraction for reference. The two composites locate different phenomena in the same parameter space.

C_R peaks at $s = 3.0$, $\lambda = 0.44$, $\sigma = 0.002$ with $C_R = 0.484$ and $f = 0.48$—precisely at the recruitment boundary (Figure 18).

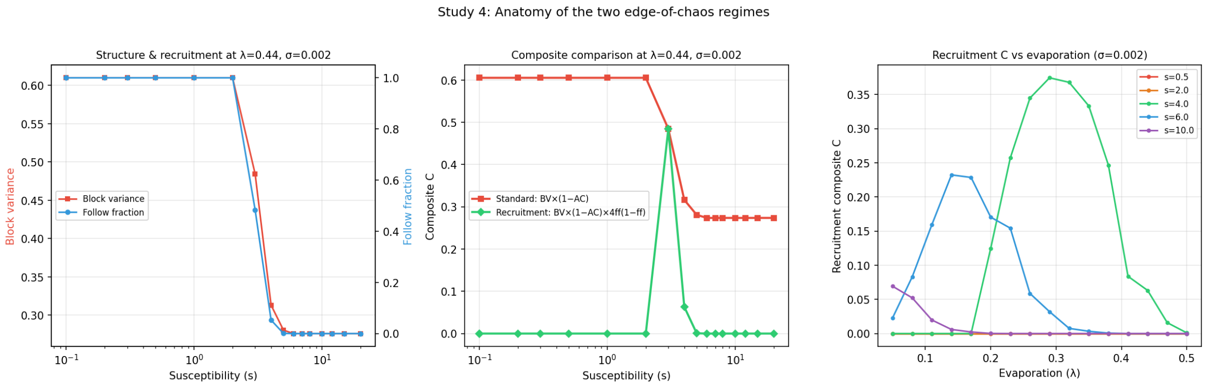

Figure 19: One-dimensional slices at the recruitment composite peak ($\lambda = 0.44$, $\sigma = 0.002$). Left: block variance and follow fraction vs susceptibility. Centre: both composites vs susceptibility—standard C drops monotonically; C_R peaks at $s \approx 3$ where $f \approx 0.5$. Right: C_R vs evaporation at several susceptibility values.

7.3 A sharp phase transition that is not edge of chaos

Across the recruitment boundary, C drops *monotonically*: at $\lambda = 0.44$, $C = 0.605$ at $s = 2$ (all following), $C = 0.485$ at $s = 3$ ($f = 0.48$), and $C = 0.316$ at $s = 4$ (no following). B and A decline together—fewer followers means weaker trails *and* less persistence. The recruitment boundary is a **sharp phase transition** ($\Delta f > 0.7$ in a single step) but **not edge of chaos**: a first-order-like collapse between attractors with no intermediate critical regime.

7.4 Erosion-driven transience: the true edge of chaos

The composite peaks at $C = 0.625$ in a different region: $s = 0.1$, $\lambda = 0.41$, $f = 1.0$. All ants follow, but evaporation erodes trails faster than they can freeze—the network is maximally structured yet constantly reshuffling. This is genuine edge of chaos along the *persistence axis*.

7.5 Why the original model's ridge was a superposition

The noise parameter ε simultaneously reduces recruitment (ants ignore pheromone) *and* persistence (disrupted deposition weakens trails), producing a single ridge. The susceptibility model separates these into orthogonal axes, revealing a superposition of erosion-driven transience (genuine edge of chaos, $C = 0.625$) and a recruitment phase transition (sharp but not edge of chaos—C drops monotonically, located by C_R at $f \approx 0.5$).

8 Study 5: Finding the Edge of Chaos

Studies 3 and 4 established that the susceptibility model's recruitment boundary is a sharp phase transition but *not* edge of chaos—the composite drops monotonically across it. The genuine edge of chaos (erosion-driven transience) operates along the persistence axis via an environmental mechanism (evaporation). Can we achieve edge of chaos through a mechanism intrinsic to the ant?

8.1 The recipe: decoupling structure from persistence

Edge of chaos requires B to stay high while A drops—trails must be strong yet transient. The susceptibility model's recruitment boundary fails because B and A decline together (only positive feedback). A mechanism that keeps trails strong while making them shift location would decouple B from A. The original model achieves this *externally* (noise) and *environmentally* (evaporation). Can an *intrinsic* ant-level mechanism do the same?

8.2 Candidate mechanisms

We test three intrinsic mechanisms, built on top of the susceptibility model:

1. **Habituation** (sensory adaptation). Each ant maintains an internal habituation counter h. While following a trail, h increments by 1 each step. When $h \geq \tau_h$ (the habituation timescale), the ant becomes desensitised: it stops following pheromone *and* stops depositing it (habituated ants contribute nothing to the trail network). Recovery is 5× faster than habituation, so the steady-state follow fraction is ∼83%. The ant resumes following only after h decays back to zero (hysteresis prevents rapid oscillation). Initial habituation counters are randomised to desynchronise the population. The control parameter is τ_h: the number of following steps before an ant habituates. Fixed susceptibility $s = 2.0$ (all-following regime).

2. **Peaked pheromone response** (sensory saturation). Rather than removing ants from trails, this mechanism modifies *how* ants perceive pheromone. The effective pheromone field used for movement decisions is $\phi_{\text{eff}} = \phi \cdot e^{-\phi/\phi_{\text{sat}}}$, which peaks at $\phi = \phi_{\text{sat}}$ and decreases above it. Ants are most attracted to moderately concentrated trails; extremely strong trails lose their attractiveness advantage. This keeps $f = 1$ (all ants always follow) but redistributes ants across competing trail segments. The control parameter is ϕ_{sat}: the pheromone concentration at peak attractiveness. Fixed susceptibility $s = 2.0$.

3. **Hill function** (soft sigmoid threshold). This replaces the hard susceptibility threshold with a biologically realistic chemoreceptor model:

$$P(\text{follow}) = \frac{\phi^n}{\phi^n + s^n} \tag{12}$$

   where ϕ is the maximum pheromone in the Moore neighbourhood, s is the half-maximum concentration, and n is the Hill coefficient (steepness). At $n = 1$, the response is hyperbolic (near-linear at low ϕ); as $n \to \infty$, it recovers the hard threshold. This creates *state-dependent noise*: strong trails are followed reliably, weak trails unreliably. Unlike the external noise parameter, this noise is intrinsic to the ant's sensory process and correlated with the pheromone field. Swept: $s \in [0.5, 20]$ and $\lambda \in [0.05, 0.50]$ at $n \in \{1, 2, 4\}$, compared against the hard threshold.

Habituation and peaked response are swept against evaporation $\lambda \in [0.05, 0.50]$ at fixed $s = 2.0$ and $\sigma = 0.005$ (10 values each axis, 100 simulations per mechanism). The Hill function sweep uses 9 susceptibility values × 10 evaporation values × 3 steepness values = 270 simulations, plus 90 hard-threshold comparisons.

8.3 Results: no mechanism produces intrinsic edge of chaos

Habituation ($\tau_h \times \lambda$, 100 simulations): The composite peaks at $\tau_h = 400$, $\lambda = 0.50$ ($C = 0.678$, $f = 0.85$)—the boundary of the swept range. Increasing τ_h monotonically increases C by keeping more ants following for longer. No intermediate τ_h produces a peak.

Peaked response ($\phi_{\text{sat}} \times \lambda$, 100 simulations): The composite peaks at $\phi_{\text{sat}} = 0$ (no modification), $\lambda = 0.45$ ($C = 0.639$, $f = 1.00$). Introducing a peaked response at any saturation level reduces the composite.

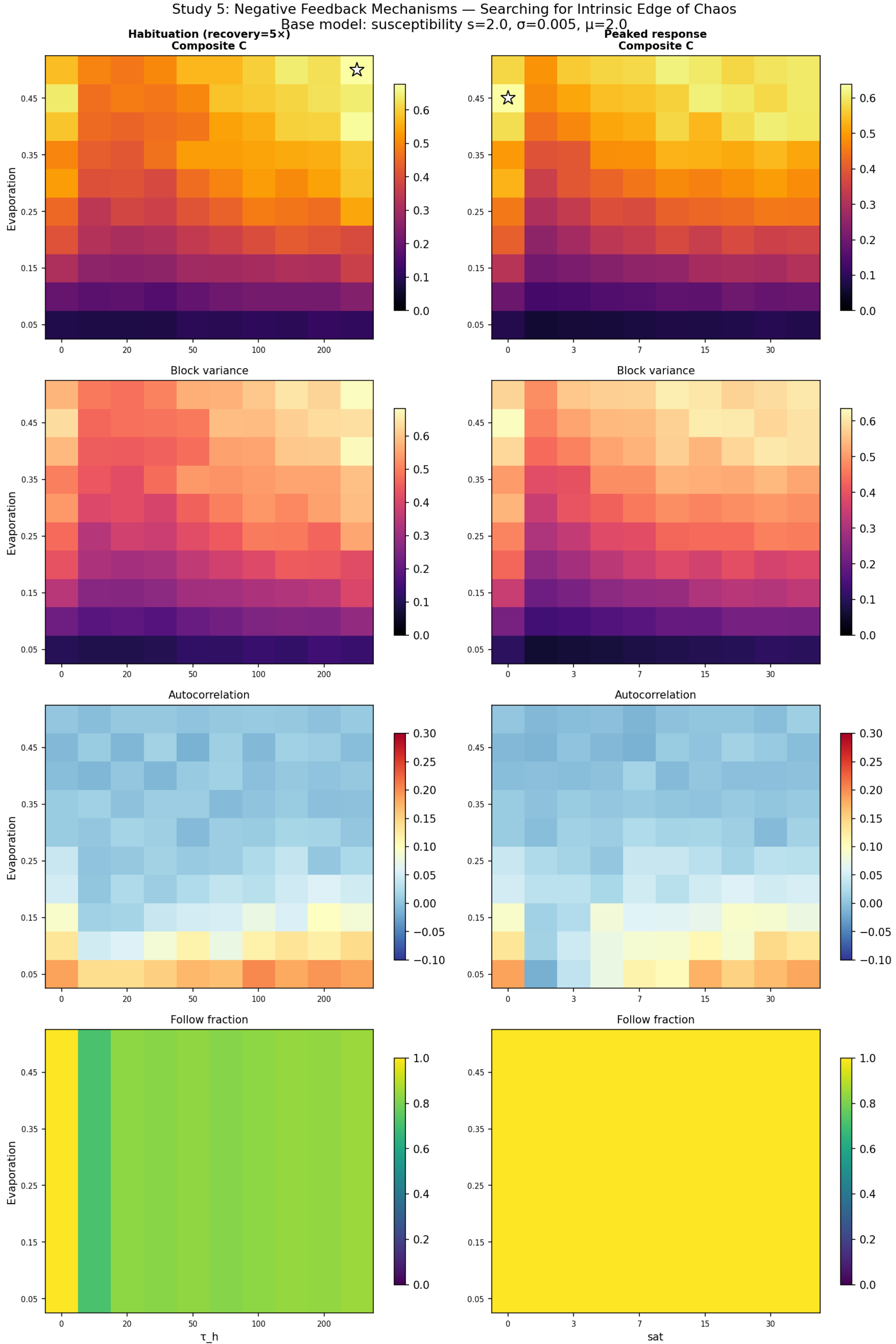

Figure 20: Composite order parameter, block variance, autocorrelation, and follow fraction for habituation and peaked response. Stars mark composite peaks, both at the boundary of the swept range. Neither produces the inverted-U signature of intrinsic edge of chaos.

Hill function ($s \times \lambda$ at $n \in \{1, 2, 4\}$, 360 simulations): The Hill function creates a smooth follow-fraction gradient—at $n = 1$, $s = 2$, $\lambda = 0.50$, the follow fraction is $f = 0.65$ rather than collapsing sharply to zero—but the composite still peaks at the lowest susceptibility ($s = 0.5$) where nearly all ants follow. At no steepness value does the composite peak at an intermediate susceptibility.

Table 4: Hill function sweep results. All composite peaks occur at $s = 0.5$ (boundary, not interior).

Threshold	Peak s	Peak λ	C	f	Interior in s?
Hard	0.5	0.45	0.639	1.00	No (boundary)
Hill n=1	0.5	0.35	0.496	0.89	No (boundary)
Hill n=2	0.5	0.45	0.597	0.98	No (boundary)
Hill n=4	0.5	0.50	0.581	1.00	No (boundary)

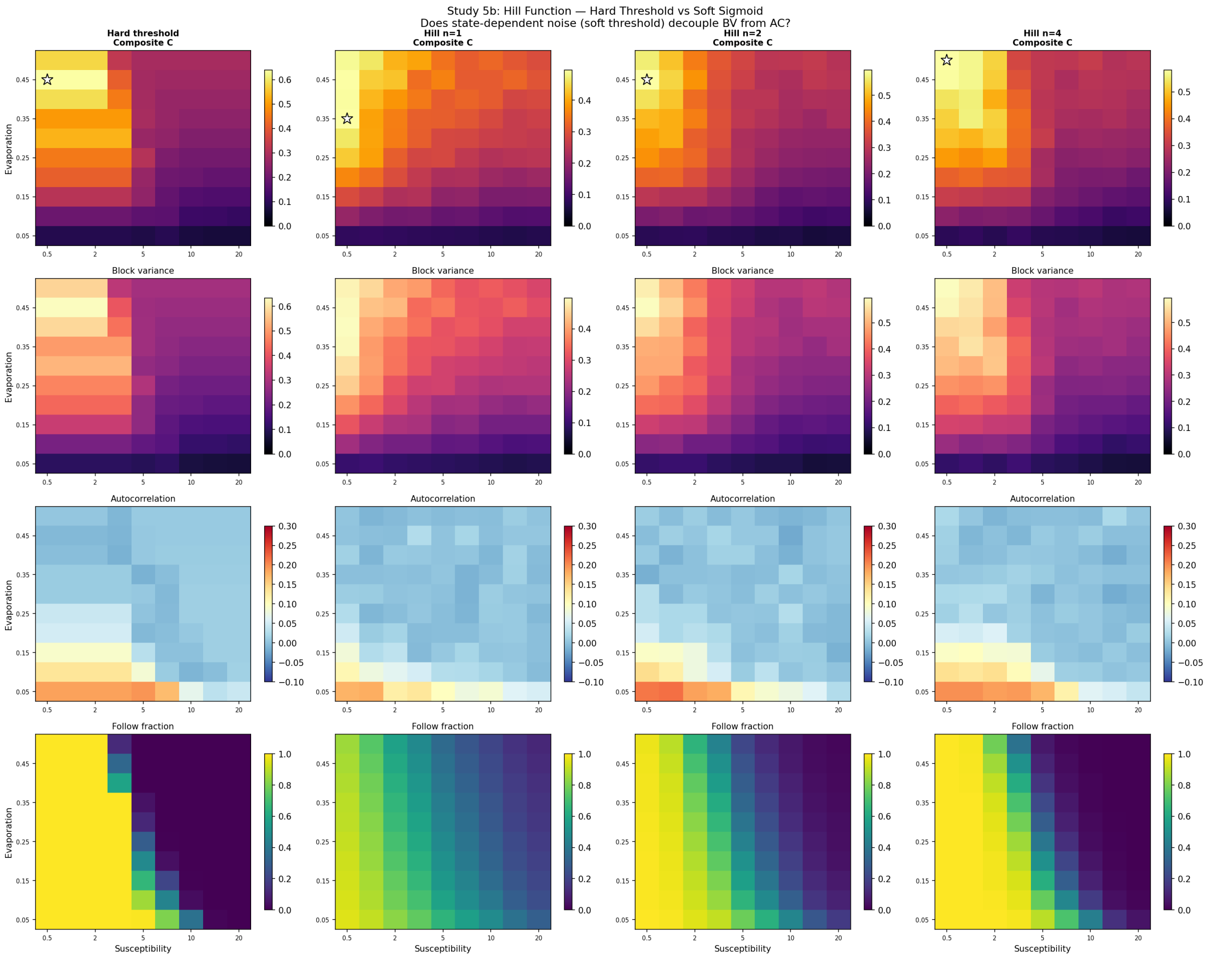

Figure 21: Hard threshold vs Hill function at three steepness values. Top row: composite C; lower rows: block variance, autocorrelation, follow fraction. The Hill function smooths the follow-fraction transition (especially at $n = 1$) but the composite always peaks at the lowest susceptibility (highest following), not at the recruitment boundary.

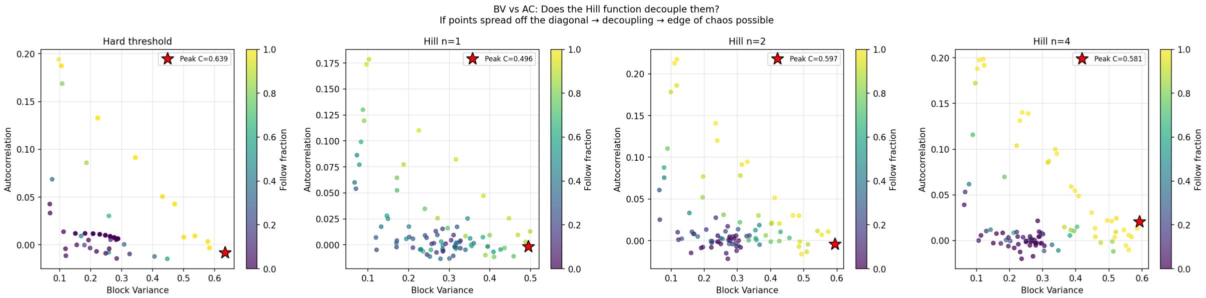

Figure 22: Block variance vs autocorrelation for each threshold type. All four show the same positive correlation between B and A: state-dependent noise does not decouple them.

8.4 Why intrinsic mechanisms fail: correlated vs uncorrelated noise

The failure of all three mechanisms reveals a structural constraint. The composite $C = B \times (1 - A)$ peaks when trails are *maximally structured yet maximally transient*—B must stay high while A drops. This requires a source of disorder that is **uncorrelated with the pheromone field**.

External noise (ε) is uncorrelated: any ant, regardless of its position relative to trails, can be randomly displaced. Strong trails lose followers at the same rate as weak trails. This creates turnover (reducing A) without specifically targeting the trails that sustain structure (preserving B).

Evaporation (λ) is similarly uncorrelated with ant behaviour: it decays the pheromone field uniformly, shifting the spatial configuration while leaving the number of followers (and thus trail strength) unchanged.

All three intrinsic mechanisms produce noise that is **correlated with the pheromone field**:

- **Habituation** removes ants from trails they have followed longest—precisely the strongest trails. This preferentially weakens the structures that contribute most to B.
- **Peaked response** redistributes ants away from the strongest trails toward weaker ones, diluting the spatial heterogeneity that B measures.
- **Hill function** creates state-dependent noise where weak trails are followed unreliably—but weak trails are the ones that *need* following to survive. The noise acts precisely where it destroys the most structure. At $n = 1$, the follow fraction smoothly interpolates from high (strong trails) to low (weak trails), but B and A still decline together because the noise-structure correlation is preserved.

In each case, the intrinsic noise is stronger where the pheromone signal is weaker, creating a positive feedback loop: weak trails suffer more noise $\rightarrow$ trails weaken further $\rightarrow$ more noise. This is the opposite of what edge of chaos requires. The composite needs a disorder source that occasionally disrupts *strong* trails (reducing A) while the system's positive feedback rebuilds structure elsewhere (maintaining B). Only uncorrelated noise—external randomness or environmental decay—achieves this.

This explains why the original noise model's edge-of-chaos ridge appeared to be intrinsic: the noise parameter ε acts *uniformly* on all ants regardless of their pheromone environment, making it functionally equivalent to an uncorrelated environmental perturbation rather than a state-dependent behavioural mechanism.

9 Discussion

The five studies reveal two principal results: (1) sharp phase transitions and edge-of-chaos dynamics are distinct phenomena, and (2) edge of chaos in stigmergic systems requires uncorrelated noise.

The susceptibility model decouples two axes of order that the original noise parameter conflated. **Recruitment** (controlled by s) determines whether ants follow trails at all—the transition is sharp (first-order-like) but B and A decline together, producing no composite peak. **Persistence** (controlled by λ) determines whether trails, once formed, survive—increasing evaporation takes the system from frozen trails through transient dynamics (peak C) to dissolution, a genuine edge-of-chaos trajectory. The original noise parameter ε traverses both axes simultaneously, which is why it produces a single clean ridge.

The susceptibility model is the most biologically realistic variant: pheromone deposition is unconditional (biochemical secretion), trail following is triggered by local concentration (olfactory detection), and the resulting phase transition emerges from collective dynamics rather than being imposed by a control parameter. But the sharp recruitment boundary is not edge of chaos. The genuine edge of chaos lies in erosion-driven transience, where environmental decay—not ant behaviour—provides the disorder.

10 Summary

Table 5: Summary of all parameter sweeps. "EoC ridge" indicates whether the composite order parameter forms a sharp peak (edge-of-chaos ridge), distinct from whether the model has a sharp phase transition in follow fraction.

Model	Sweep axes	Struct. metric	Peak C	EoC ridge?	Mechanism
Original	$\varepsilon \times \lambda$	Linearity	0.312	Yes	Coupled destruction
Simple (μ=3)	$p_f \times \lambda$	Linearity	0.309	No	Decoupled control
Simple (μ=0.5)	$p_f \times \lambda$	Linearity	0.287	No	Decoupled control
Simple (3D)	$p_f \times \lambda \times \sigma$	Linearity	0.300	No	Decoupled control
Suscept. (v2)	$\boldsymbol{s \times \lambda \times \sigma}$	**Block var.**	**0.625**	**Yes**†	**Erosion-driven**
Suscept. (C_R)	$s \times \lambda \times \sigma$	Block var. $\times 4f(1-f)$	0.484	No‡	Recruitment collapse
Habituation	$\tau_h \times \lambda$	Block var.	0.678§	No	Correlated neg. feedback
Peaked resp.	$\phi_{\text{sat}} \times \lambda$	Block var.	0.639§	No	Correlated redistribution
Hill (n=2)	$s \times \lambda$	Block var.	0.597§	No	Correlated state-dep. noise

†Along persistence axis (evaporation), not at recruitment boundary.
‡Sharp phase transition in f, but composite drops monotonically—not edge of chaos.
§Peak at boundary of swept range—no interior peak in the intrinsic parameter.

The progression from original to simple to susceptibility model traces a path toward biological realism. The **original model** produces edge of chaos but through an artificial noise parameter. The **simple model** shows that replacing noise with a follow probability is insufficient: p_f produces a smooth crossover because it only controls signal reading, not writing, and momentum creates structure independently. The **susceptibility model** achieves a sharp recruitment phase transition through positive feedback—but this is a first-order collapse, not edge of chaos. The **decoupling analysis** (Study 4) reveals that the original model's ridge was a superposition of two phenomena: a recruitment collapse and erosion-driven transience; only the latter is edge of chaos (C = 0.625). The **intrinsic mechanism experiments** (Study 5) test habituation, peaked response, and Hill function thresholds; all fail because their noise is correlated with the pheromone field.

The central finding is a necessary condition for edge of chaos in stigmergic systems:

> **Edge of chaos requires a source of disorder that is uncorrelated with the collective signal.**

$C = B \times (1 - A)$ peaks only when B and A are decoupled. Any disorder source whose intensity depends on the pheromone field couples them: noise targeting weak trails (Hill function, susceptibility threshold) collapses structure; noise targeting strong trails (habituation) directly weakens the dominant features. Only disorder that acts *independently* of the field can occasionally disrupt strong trails (reducing A) while positive feedback rebuilds structure elsewhere (maintaining B).

We tested this against five intrinsic mechanisms—follow probability, susceptibility threshold, habituation, sensory saturation, and Hill function chemoreception—spanning qualitatively different approaches to behavioural disorder. All five produce noise correlated with the pheromone field; none produces an edge-of-chaos ridge. The two mechanisms that succeed—external noise (ε) and environmental decay (λ)—both act uniformly, independent of the current field configuration.

This reframes the original noise model: ε succeeds not because it is "intrinsic to the ant" but because it is *uncorrelated* with the pheromone field. An ant that moves randomly with probability ε, regardless of what it smells, is functionally equivalent to an environmental perturbation. A biologically realistic mechanism with the same property (e.g., motor noise from stochastic neural firing, independent of olfactory input) would produce edge of chaos—but would be indistinguishable in the model from the external noise it replaces.